ISO/IEC 25010 Software Evaluation

- **Project Name:** Makikibahay: A Web-Based Boarding House Finder with Smart Filtering and a 3D Room Preview for Cabanatuan City
- **Evaluation Date:** [19/02/2026]
- **Evaluator(s):** [Names and Roles]
- **Project Description:**
  Makikibahay is a web-based boarding house finder for Cabanatuan City that features smart filtering and 3D room previews. It is a full-stack, location-aware platform that personalizes results based on user type, budget, and proximity to schools or workplaces. The system helps users make informed decisions while giving property owners better control over their online listings.

  The app has main features:

  - **Centralized Verified Listings** – A structured database that gathers all boarding house listings in Cabanatuan City, complete with verified details such as pricing, amenities, house rules, and photos to ensure accuracy and transparency.
  - **Smart Filtering & Personalized Recommendations** – Advanced filtering and hybrid recommendation system based on user type (student or worker), budget range, preferred amenities, and proximity to schools or workplaces.
  - **3D Room Preview Interactive** – 3D visualization that allows users to virtually explore rooms before visiting, reducing unnecessary travel and helping tenants make informed decisions.
  - **Landlord Dashboard & Management Tools** – A dedicated dashboard for property owners to manage listings, update availability, track inquiries, and communicate with potential tenants efficiently.

Evaluation Criteria

1. **Usability (User-Based Evaluation)** (see page 4 for more details)
   Objective: Assess how easy and effective the system is for students and guests to use.

Sub-characteristic	Metric	Evaluation Method	Findings
Learnability	Time taken to complete common tasks (e.g., searching for a building)	User testing (time tracking)	[Enter Findings]
Operability	User satisfaction score (1-10)	User survey	[Enter Findings]
Accessibility	Compliance with WCAG standards	Accessibility testing tools	[Enter Findings]
Error Prevention	Number of user errors recorded	Usability testing reports	[Enter Findings]

   Evidence: Screenshots, feedback forms, video recordings of user tests.

2. **Functionality (Expert/Developer Evaluation with Logs)** (see page 12 for more details)
   Objective: Verify that all features work correctly and meet requirements.

Sub-characteristic	Metric	Evaluation Method	Findings
Functional Completeness	% of required features implemented	Requirement traceability matrix	[Enter Findings]

Functional Correctness	% of correctly working features	Functional testing logs	[Enter Findings]
Functional Appropriateness	Task to complete tasks successfully	User testing	[Enter Findings]

Evidence: System logs, screenshots, test case results.

3. **Reliability (Experimental Evaluation)** (see page 8 for more details)
   Objective: Ensure the system remains stable and error-free.

Sub-characteristic	Metric	Evaluation Method	Findings
Maturity	Mean Time Between Failures (MTBF)	System logs & failure tracking	[Enter Findings]
Availability	% uptime during testing	System performance monitoring	[Enter Findings]
Fault Tolerance	Error recovery time	Fault injection testing	[Enter Findings]
Recoverability	Data recovery success rate	Backup & restore testing	[Enter Findings]

Evidence: System uptime reports, error logs, recovery test results.

4. **Accuracy** (see page 16 for more details)
   Objective: Verify the correctness of displayed data.

Sub-characteristic	Metric	Evaluation Method	Findings
Data Accuracy	% of correct information displayed	Manual verification	[Enter Findings]
GPS Accuracy	Location deviation (in meters)	GPS testing tools	[Enter Findings]

Evidence: Screenshots, GPS test results, error logs

Summary & Recommendations

Overall Score/Findings: [Provide a summary of evaluation results]
Key Strengths: [List strengths]
Areas for Improvement: [List areas requiring improvement]
Recommendations: [Suggest optimization steps]

Approval & Sign-off

- **Evaluator Name & Signature:** [Name & Signature]
- **Project Manager Name & Signature:** [Name & Signature]
- **Date:** [19/02/2026]

– End of Evaluation Report –

Usability Evaluation Methods

- **User Testing (Time Tracking)**

  Objective: Measure how long it takes users to complete common tasks to assess ease of use and efficiency.

  Procedure:

  1. Select **test participants** (students, workers, and potential tenants).
  2. Define **test scenarios**, such as:
     - Searching for a boarding house within a specific budget range.
     - Filtering boarding houses near a specific school or workplace.
     - Opening a listing and viewing the 3D room preview.
     - Checking detailed information (price, amenities, availability).
     - Completing the onboarding questionnaire to receive recommendations.
  3. Ask participants to complete each task while recording the time taken.
  4. Collect and analyze data to identify usability issues.
  5. Collect and analyze the recorded time data to identify usability issues and areas for improvement.

Data Collection (Feature: Smart Filtering & Search):

Task	User ID	Time Taken (Seconds)	Completed Successfully? (Yes/No)	Issues Encountered
Use filters to find a boarding house within ₱3,000–₱5,000 budget.	User 1			
Search for a boarding house near a specific school (e.g., within 1 km).	User 2			
Apply filters for preferred amenities (WiFi, own CR, kitchen access).	User 3			
Complete the onboarding questionnaire and review recommended listings.	User 4			

Data Collection (Feature: 3D Room Preview):

Task	User ID	Time Taken (Seconds)	Completed Successfully? (Yes/No)	Issues Encountered
Open a boarding house listing and launch the 3D room preview.	User 1			

Navigate inside the 3D room (rotate, zoom, and explore).	User 2			
Identify key room features (bed area, comfort room, storage space).	User 3			
Exit the 3D preview and return to the listing page.	User 4			

Data Collection (Feature: Verified Listings & Reviews):

Task	User ID	Time Taken (Seconds)	Completed Successfully? (Yes/No)	Issues Encountered
View full details of a boarding house listing.	User 1			
Check pricing, amenities, and house rules.	User 2			
Read tenant reviews and ratings.	User 3			
Verify room availability status.	User 4			

Data Collection (Feature: Landlord Dashboard):

Task	User ID	Time Taken (Seconds)	Completed Successfully? (Yes/No)	Issues Encountered
Log in as a property owner.	User 1			
Add a new boarding house listing.	User 2			
Update room availability status.	User 3			
Respond to a tenant inquiry message.	User 4			

Evidence:

- **User Survey**

  Objective: Gather feedback on user experience, satisfaction, and ease of use.

  Survey Questions (Scale: 1 - Strongly Disagree, 4 - Strongly Agree)

Question	1	2	3	4
The platform is easy to use.				
The search and filtering features are effective.				
A 3D room preview would reduce unnecessary visits.				
It is easy to find suitable boarding houses.				
The listing information is clear and complete.				
The recommendations match my preferences.				
The design and layout are visually appealing.				
I feel confident using this platform to choose a boarding house.				

  Additional Open-Ended Questions:

  1. What features of Makikibahay did you find most useful?
  2. What improvements or additional features would you suggest?
  3. Did you encounter any difficulties while using the platform? If yes, please describe them.

  Evidence:

- **Accessibility Testing Tools**

  Objective: Ensure the app is accessible to all users, including those with disabilities.

  Testing Tools:

  - **Google Lighthouse** (Accessibility Audit)
  - **WAVE Web Accessibility Evaluation Tool** (Checks for WCAG compliance)
  - **axe DevTools** (Detects accessibility issues in UI elements)
  - **NVDA Screen Reader** (Simulates navigation for visually impaired users)

  Evaluation Process:

  1. **Run Google Lighthouse** to check for testing accessibility (e.g., missing alt text, poor contrast, and keyboard navigation issues).

     Data Collection Format:

Issue Detected	Severity	Tool Used	Suggested Fix

  Evidence:

Usability Testing Reports

Objective: Summarize findings from usability evaluations, highlighting strengths and areas for improvement.

Introduction

[Purpose and scope of the usability test.]

Methodology [Description of user testing, surveys, and accessibility evaluation.]

Findings [Summary of usability scores, key user challenges, and accessibility issues.]

Recommendations [Proposed improvements based on test results.]

Supporting Evidence [Screenshots, logs, survey graphs, and recorded test sessions.]

– End of Usability Evaluation Methods –

Reliability Evaluation Methods

- **System Logs & Failure Tracking**

  Objective: Track system crashes, errors, and performance issues to measure reliability.

  Procedure:

  1. Enable logging in Laravel (`storage/logs/laravel.log`).
  2. Use monitoring tools (e.g., Laravel Telescope, Sentry, or LogRocket) to collect error logs.
  3. Analyze logs for common errors, frequency of crashes, and error trends.
  4. Calculate Mean Time Between Failures (MTBF):

$$MTBF = \frac{Total\ Uptime}{Number\ of\ Failures}$$

  Data Collection Format:

Date/Time	Error Type	Error Message	Affected Module	Occurrence Count	Resolution Status

  Evidence:

- **System Performance Monitoring**

  Objective: Ensure the app maintains stable performance under expected conditions.

  Procedure:

  1. Use performance monitoring tools like:
     - Laravel Debugbar (tracks database queries, and load times).
     - Lighthouse (measures page performance).
     - New Relic or Datadog (monitors real-time server performance).
  2. Measure key metrics:
     - Response time (how fast the system reacts).
     - CPU & memory usage (how efficiently resources are used).
     - Server uptime (percentage of time the app is running).

Data Collection Format:

Metric	Expected Value	Actual Value	Pass/Fail	Remarks
Page Load Time	< 2 sec			
Time to retrieve announcements				
Response time for common users				
API Response Time	< 500ms			
Memory Usage	< 70% of server capacity			
Uptime	99.9%			

Evidence:

- **Fault Injection Testing**

  Objective: Test system stability by simulating unexpected failures.

  Procedure:

  1. Introduce controlled failures in a test environment:
     - Kill database connection to check how the system handles downtime.
     - Overload the system with heavy traffic (e.g., Apache JMeter stress test).
     - Simulate API failures to see if error-handling mechanisms work.
  2. Monitor the system's response and recovery time.
  3. Log errors and check if automatic recovery mechanisms work.

Data Collection Format:

Fault Injected	Expected Behavior	Actual Behavior	Recovery Time	Pass/Fail
Database Failure	Show user-friendly error message			
High Traffic (1000 users)	Maintain stable performance			
API Timeout	Retry API request automatically			

Evidence:

- **Backup & Restore Testing**

Objective: Verify that data can be successfully backed up and restored after a failure.

Procedure:

1. Schedule automated database backups (e.g., via Laravel backup package).
2. Simulate a data loss event (e.g., delete records from the database).
3. Restore the latest backup and verify data integrity.
4. Check if all functionalities work properly after restoration.

Data Collection Format:

Backup Date	Backup Size	Data Loss Simulated? (Yes/No)	Restore Success? (Yes/No)	Time Taken to Restore	Pass/Fail

Evidence:

– End of Reliability Evaluation Methods –

Functionality Evaluation Methods

- **Requirement Traceability Matrix (RTM)**

  Objective: Ensure that all functional requirements are implemented and tested.

  Structure:

Requirement ID	Requirement Description	Module/Feature	Test Case ID	Test Status (Pass/Fail)	Remarks
FR-001	Display verified boarding house listings with complete details (price, amenities, rules, photos).	Listings Module	TC-001		
FR-002	Allow users to filter and search boarding houses based on budget, amenities, and location.	Smart Filtering & Search	TC-002		
FR-003	Provide personalized recommendations based on user profile and preferences.	Recommendation System	TC-003		
FR-004	Requirement Description: Enable users to view interactive 3D room previews.	3D Room Preview	TC-004		
FR-005	Allow property owners to add, edit, and manage boarding house listings.	Landlord Dashboard	TC-005		

  Steps to Create RTM:

  1. List **all functional requirements** from the system specifications.
  2. Map each requirement to a **specific feature/module** in the app.
  3. Link the requirement to a **test case ID** that verifies it.
  4. Record the test **status** (Pass/Fail).
  5. Add **remarks** for issues or improvements.

  Evidence:

- **Functional Testing Logs**

  Objective: Verify that features work correctly under different conditions.

  Structure:

Test Case ID	Feature	Test Steps	Expected Result	Actual Result	Status (Pass/Fail)	Remarks
TC-001	Listings Module	Verify that property information is displayed.	The listing and its details are displayed correctly without errors.			
TC-002	Smart Filtering & Search	Verify that filtering and search return accurate results.	Filters work in combination without errors.			
TC-003	Recommendation System	Verify that the recommendation system suggests relevant properties.	Recommended properties reflect user preferences.			
TC-004	3D Room Preview	Navigate to a property that offers a 3D room preview.	3D preview loads properly, is interactive, and visually matches property details.			
TC-005	Landlord Dashboard	Perform add/edit actions on property listings.	Listings are added/edited correctly, and dashboard data updates in real time.			

  Steps for Functional Testing:

  1. Create **test cases** for each feature.
  2. Document the **test steps** and expected behavior.
  3. Run tests and record **actual results**.
  4. Mark each test as **Pass/Fail**.
  5. Attach **logs/screenshots** as proof.

  Evidence:

- **User Testing (Functionality Check by Real Users)**

Objective: Validate that Makikibahay functions correctly for real users in practical scenarios, ensuring all features work as intended.

Procedure:

1. **Recruit users** (students, guests, faculty).
2. Assign **specific tasks to test different features**, such as:
   - Search for a boarding house near a specific school.
   - Use filters to find a boarding house within a certain budget.
   - Open a listing and view the 3D room preview.
   - Check listing details including price, amenities, and availability.
   - Add a new listing or update an existing one (for landlord users).
   - Navigate to a boarding house location using the map.
   - Read tenant reviews for a specific boarding house.
3. Observe and record **user interactions**.
4. Collect feedback on any functionality issues or suggestions for improvement.

Data Collection Format:

Task	User ID	Task Completed Successfully? (Yes/No)	Errors Encountered	Suggestions
Use filters to find a boarding house within a certain budget	User 1			
Open a listing and view 3D room preview	User 2			
Check listing details (price, amenities, availability)	User 3			
Add or update a boarding house listing (landlord)	User 4			
Navigate to a boarding house using the map	User 5			
Read tenant reviews for a specific boarding house	User 6			

Evidence:

– End of Functionality Evaluation Methods –

Accuracy Evaluation Methods

- **Manual Verification**

  Objective: Ensure the correctness and reliability of boarding house listings, 3D previews, landlord details, and location data displayed in Makikibahay.

  Procedure:

  1. **Verify Boarding House Listings**
     - Cross-check listing details (price, amenities, rules, availability) with the property owner.
     - Confirm that uploaded photos match the actual boarding house.
     - Ensure listings are properly categorized and tagged (e.g., student-friendly, near school, with WiFi).
  2. **Verify 3D Room Preview**
     - Compare the 3D room model with actual room photos or on-site inspection.
     - Confirm that room layout, furniture placement, and dimensions are accurate.
  3. **Verify Location & Distance Data**
     - Compare mapped boarding house locations with Google Maps or actual site coordinates.
     - Ensure calculated distance from schools/offices is accurate.
  4. **Verify Reviews & Availability Status**
     - Confirm that reviews are submitted by registered users only.
     - Check that room availability status matches the landlord's dashboard records.

Data Collection Format:

Item Verified	Source of Truth	App Data Matches? (Yes/No)	Errors Found	Corrections Needed?
Boarding house price and amenities	Direct confirmation from landlord			
Room availability status	Landlord dashboard record			
3D room layout	Actual room photos / site visit			
Boarding house location pin	Google Maps coordinates			
Tenant reviews	Registered user submissions			
Listing photos	On-site verification / landlord-provided images			

Evidence:

- **GPS Testing Tools**

  Objective: Measure the accuracy of GPS-based navigation and campus positioning.

  Tools Used:

  - **Google Maps** – To verify the coordinates of buildings and key locations.
  - **Geo Tracker** – To track movement and check route accuracy.

  Procedure:

  1. **Test location accuracy**
     - Compare the app's **displayed coordinates** of buildings and landmarks with **Google Maps data**.
     - Measure the **deviation in meters** between the actual location and the app's pin position.
  2. **Test route accuracy**
     - Use the app's **navigation feature** to generate routes between two points.
     - Walk the suggested route and check for **incorrect paths or missing roads**.

  Data Collection Format:

Location	Actual Coordinates (Google Maps)	App Coordinates	Deviation (Meters)	Pass/Fail
Main Gate				
Registrar's Office				
University Library				
University Gymnasium				

  Evidence:

  - Screenshots of Google Maps comparisons.
  - Recorded test routes using GPS tracking apps.

– End of Accuracy Evaluation Methods –